

Teacher policies to support the use of digital resources in the classroom



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Education Policy Perspectives

Teacher policies to support the use of digital resources in the classroom

Abstract

The digital transformation of education presents both opportunities and challenges for teaching practices and teachers' working conditions. While many educators have strengthened their digital competencies in recent years, persistent technical and pedagogical barriers continue to hinder the effective and deliberate integration of digital tools in the classroom. This policy paper was prepared as part of the OECD project "Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers". It examines how teacher policies can be adapted to better support the profession in navigating this transition. Drawing on insights from PISA 2022 and comparative data from the Policy Survey on School Education in the Digital Age, it addresses the following questions: What forms of central guidance can support teachers' effective use of digital resources? To what extent do professional standards and appraisal frameworks reflect the evolving digital education landscape? How can technology alleviate teachers' workload and optimise their use of time? Do staff roles and career structures need to be reshaped to enhance schools' digital capacity? Which mechanisms can ensure that teachers receive ongoing technical assistance and support in maintaining digital resources? The paper offers evidence-informed insights for policy makers seeking to strengthen teacher policies to build resilient, future-ready digital education systems.

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Executive summary

The digital transformation of education has profound implications for the teaching profession. The COVID-19 pandemic has contributed to a significant increase in the use of digital education resources and teachers have greatly improved their capacity to use technology to enhance teaching and learning. Nevertheless, both technical and pedagogical challenges concerning the effective and deliberate use of digital resources remain. As technology transforms teaching and learning in schools, teacher policies may need to be adapted to provide continued support to the profession and to address the new challenges that technology introduces in the classroom. Drawing on insights from PISA 2022 and comparative data collected through the Policy Survey on School Education in the Digital Age, this paper presents evidence and policy examples from high-performing education systems to address the following questions:

1. What forms of central guidance can support teachers' effective work with digital resources?
2. Do professional standards and appraisal frameworks reflect the digital transformation of teaching?
3. How can technology ease teachers' workload and help them to use their time more effectively?
4. Which staff roles and career structures can support schools in their digital transformation?
5. How can teachers be provided with technical assistance and support in maintaining digital resources?

Education authorities have recognised the important implications that digitalisation will have for their ability to attract, retain and develop teachers in the coming years. Results of the Policy Survey indicate that many education systems have already introduced reforms and initiatives aimed at supporting innovative practices and the effective and deliberate use of digital technologies in schools. For example, nearly all responding jurisdictions provide some form of central guidance and support to promote teachers' effective use of digital resources. Such initiatives frequently involve the development of online platforms that allow teachers to access guidance on the effective use of digital resources or to share digital education content and good practices with their peers. Around half of the participating jurisdictions have also launched initiatives specifically focused on the use of artificial intelligence (AI) for teaching.

Professional standards and appraisal frameworks can be another powerful tool to set expectations for high-quality teaching and to guide educators to attain the skills they need to be successful in the digital age. Most jurisdictions participating in the Policy Survey now address the use of digital resources in teachers' professional standards. To ensure that these standards evolve and remain relevant amid rapid technological change, some education systems have engaged in structured reflections with stakeholders to explore how teachers' roles may evolve in the future. By contrast, only half of the jurisdictions systematically consider teachers' work with digital resources as part of their regular appraisal.

Digital tools have the potential to make teachers' work more efficient but also require them to spend time to learn about and evaluate new digital technologies and to adapt their teaching practices accordingly. Setting aside time for teachers to work with digital resources can help them to make informed decisions on their integration in the classroom. However, it is also important to consider how digital technologies can reduce the time that teachers spend on repetitive or administrative tasks. Some countries have used this strategy successfully to ease teachers' workload and to help them focus on improving students' learning outcomes.

Over the past decades, many school systems have created formal roles to foster innovation in schools and to support the effective use of digital technologies. This has allowed schools to take a more strategic approach to digital education and created new opportunities for teachers' professional growth. Schools or local authorities tend to be in charge of filling these positions and allocating responsibilities for both the technical and the pedagogical aspects of digital education in schools. Yet, central authorities play an important role in defining these positions and their responsibilities. Given the complex combination of skills and expertise required to successfully integrate digital resources in schools, several countries have recently promoted team-based approaches to address this challenge or fostered co-operation between staff at different levels of the system to strengthen the ICT support for schools.

1. Introduction

This is the second in a series of policy papers developed as part of the OECD project “Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers” (OECD, 2025^[1]). The project investigates how countries can design system-level policies that enable their school sectors to achieve excellence and equity in the digital age. The policy papers draw on insights from the project’s Policy Survey on School Education in the Digital Age (Boeskens and Meyer, 2025^[2]) and additional data sources to provide a more in-depth discussion of evidence and country practices related to selected policy issues covered by the project’s analytical framework (see Annex A). The first policy paper of the series focused on teachers’ continuing professional learning (OECD, 2025^[3]). This second policy paper focuses on the adaptation of other teacher policies for the digital age and their potential to support the effective and deliberate use of digital technologies in schools, highlighting practices observed in high-performing education systems.

2. Helping teachers to effectively navigate the integration of digital resources in the classroom

The use of digital technology in schools has increased considerably in recent years and OECD countries have made significant investments in their digital infrastructure. Education authorities and stakeholders alike recognise that this trend has profound implications for the teaching profession. In 2024, 88% of the education ministries responding to the OECD Education Policy Outlook Survey expected digitalisation to be of high or very high importance for attracting, retaining and nurturing teachers in 2025-30 (surpassed only by demographic changes in its importance among global trends) (OECD, 2024, p. 30^[4]). 84% of the responding ministries also reported that they were already taking policy actions to address this trend (OECD, 2024, p. 31^[4]).

These efforts are reflected in teachers' increasing sense of preparedness to use digital technologies. Data from the OECD Programme for International Student Assessment (PISA) show that, in 2022, 88% of 15-year-old students attended schools whose principal believed that their teachers had the necessary technical and pedagogical skills to integrate digital devices in instruction. This marked a 24-percentage point increase across OECD countries since 2018 (OECD, 2023, pp. 409, Table II.B1.5.32^[5]). Over the same period, the proportion of students whose principals agreed that teachers had effective professional resources to learn how to use digital devices increased from 64% to 76% (OECD, 2023, pp. 409, Table II.B1.5.32^[5]).

Nevertheless, the effective integration of digital resources continues to present teachers with an evolving set of challenges and evidence suggests that the potential of digital resources to enable more personalised and interactive learning remains far from realised (OECD, forthcoming^[6]; OECD, 2023^[7]). Many teachers find it difficult to navigate and select digital learning tools, to strike the right balance between digital and analogue resources, and to integrate them effectively. At the same time, teachers face new challenges, such as cyberbullying, cognitive overload and digital distractions in the classroom, as well as risks related to data privacy and information security (Astatke, Weng and Chen, 2023^[8]; Olsson and Granberg, 2024^[9]; OECD, forthcoming^[10]). Working with complex technologies, such as augmented reality or AI-driven feedback and learning systems, also requires teachers to have advanced digital competencies and the ability to weigh both pedagogical and ethical implications when introducing them in the classroom (Pellas, Kazanidis and Palaigeorgiou, 2020^[11]; Martínez-Comesaña et al., 2023^[12]; Chiu et al., 2024^[13]).

In this context, it is critical to design policies that enable educators to create engaging learning environments that balance cognitive demands, that effectively integrate digital and analogue methods, and that remain aligned with broader curricular goals (Olsson and Granberg, 2024^[9]; Amdam et al., 2024^[14]; Hillmayr et al., 2020^[15]). To meet these expectations, teachers must not only develop new technical and pedagogical competencies but also take on expanded roles as facilitators, guides and providers of emotional support for students navigating digital resources (OECD, forthcoming^[10]). Well-designed teacher policies can play a role in helping teachers to meet these new demands and to overcome the challenges they may face when integrating digital resources effectively into their practice.

3. Strategic planning and guidance to support teachers' effective and deliberate use of digital resources

Many countries address teachers' use of digital resources in central strategies and require schools to develop corresponding plans

Given the central role of teachers in bringing about the digital transformation of schools, many OECD countries have made it a strategic priority to strengthen teachers' capacity for digital education and to promote their effective use of digital technologies (Rosa Alessia and Taddeo Gabriella, 2022^[16]; OECD, 2019^[17]; Rizza, 2011^[18]). Results of the Policy Survey on School Education in the Digital Age (hereafter "the Policy Survey", see Box 1) show that the use of digital resources for teaching is mentioned in all but one of the 32 central-level strategies guiding digital education policy in the participating jurisdictions. Of these strategies, 7 include specific time-bound goals on the use of digital resources for teaching and 19 describe specific policy initiatives on the subject (Boeskens and Meyer, 2025, pp. 13, Figure 1.2^[2]).

Box 1. The Policy Survey on School Education in the Digital Age

The Policy Survey on School Education in the Digital Age was administered to collect international comparative evidence on digital school education policies. The survey was completed by 37 jurisdictions (26 OECD member countries, 7 sub-national entities and 4 non-member economies).¹ The Policy Survey and its associated outputs were designed to inform the OECD project "Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers". The Policy Survey was administered over the course of 2024 and reflects the policies of participating jurisdictions in January 2025. The Policy Survey covers primary and secondary education (ISCED 1-3) and its complete results are presented in Boeskens and Meyer (2025^[2]).

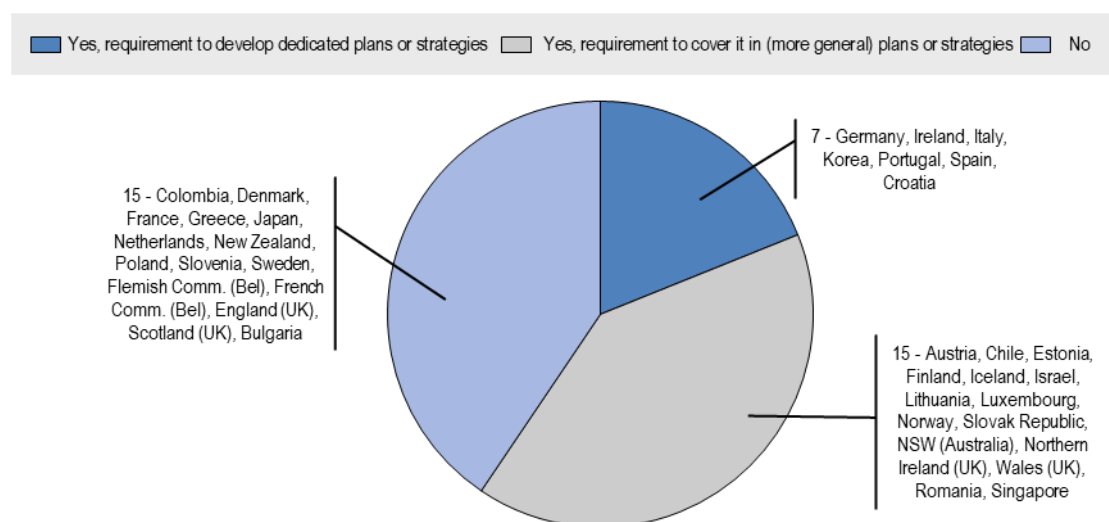
Source: Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>.

For central-level strategies to lead to meaningful changes in the classroom, they need to be translated into practical actions at the school level (OECD, 2020^[19]; Viennet and Pont, 2017^[20]). In line with this principle, most jurisdictions participating in the Policy Survey require schools to engage in strategic reflections on their approach to digitalisation. 15 of 37 systems – including high-performing systems, such as Estonia, Finland and Singapore – require schools to cover the use of digital resources for teaching in the general plans or strategies that they develop. Another 7 systems require schools to develop plans or strategies that are specifically dedicated to digital education (Figure 1). The extent to which such school-level

strategies can support teachers in the effective use of digital technologies will vary in practice, but they can be an important lever to align school-level practices with system-wide objectives and stimulate reflections on how they can be applied to each school's specific context (OECD, 2023^[21]).

Figure 1. School-level strategies for digital education

Number of jurisdictions that require schools to develop plans or strategies concerning the use of digital technology for teaching



Note: Number of jurisdictions with available data = 37. See Annex Table B.1 for further details and individual jurisdictions' responses corresponding to this figure.

Source: Reproduced from Boeskens, L. and K. Meyer (2025^[21]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Figure 1.4.

Many schools have written statements on the use of digital resources, but pedagogical guidelines for digital education are less common

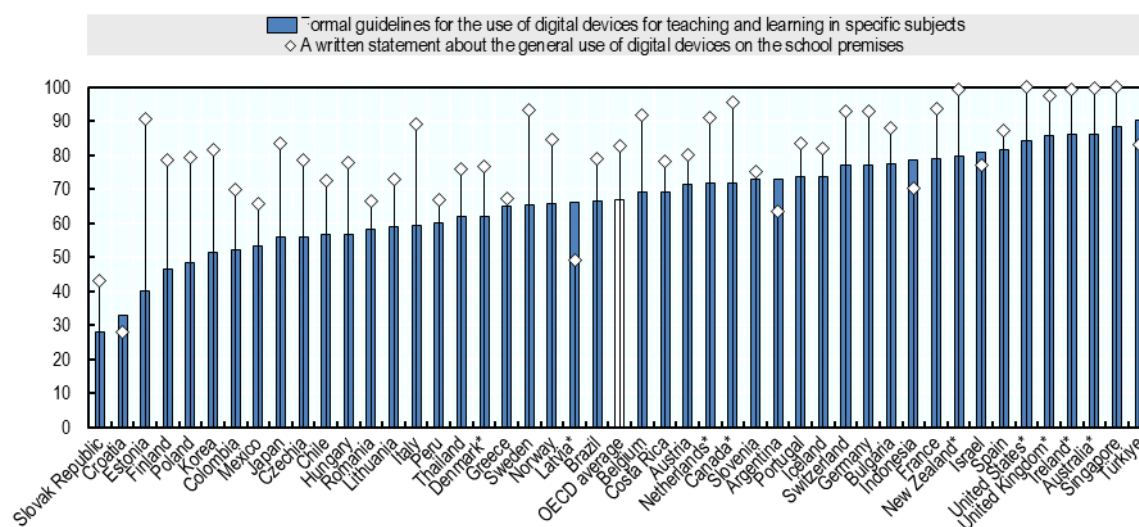
Issuing guidelines on the use of digital resources can be a means to spread best practices, to steer different actors' engagement with education technologies and to promote their safe and effective use in the classroom (OECD, 2023^[7]). In PISA 2022, 67% of 15-year-old students across OECD countries attended schools with pedagogical guidelines that explicitly addressed the use of digital devices for teaching and learning in specific subjects (Figure 2). The presence of such subject-specific guidelines for digital education was positively and significantly associated with schools' preparedness for digital learning in 34 of 81 countries/economies participating in PISA 2022, and across OECD countries, after accounting for the number of computers per student (OECD, 2023, pp. 184, Figure II.5.8^[5]).

It was even more common for students to attend schools that had a written statement about the general use of digital devices on school premises (rather than their subject-specific use), with 83% of students doing so across OECD countries in PISA 2022. Finland and Estonia were among the countries where general statements about the use of digital devices were widespread, whereas guidelines on their use for teaching and learning in specific subjects were less common (Figure 2). This highlights that subject-oriented guidelines are only one of many means by which teachers can be supported in integrating digital tools in their practice. Particularly in the context of digital education, some systems have sought to cultivate experimentation and flexibility, which may be served by strengthening teachers' professionalism and their

general technological pedagogical content knowledge (TPACK) as much as by formal guidance on specific subjects (Conrads et al., 2017^[22]; Willermark, 2018^[23]).

Figure 2. School guidelines on the use of digital devices (2022)

Percentage of 15-year-old students whose school leader reported that their school has the following



Note: Entities marked with an asterisk (*) did not meet one or more PISA sampling standards and it is not possible to exclude the possibility of more than minimal bias based on the information available at the time of data adjudication.

Source: OECD (2023^[5]), *PISA 2022 Results (Volume II): Learning During – and From – Disruption*, <https://doi.org/10.1787/a97db61c-en>, Table II.B1.5.35.

Guidance on digital education resources can be issued at the central level or developed locally. It can cover both technical and pedagogical matters and can be directed at a range of audiences, including teachers, students and other stakeholders. To support teachers in navigating new digital technologies in a fast-evolving field, guidance on the use of digital tools needs to be up to date (European Commission, 2023^[24]; UNESCO, 2018^[25]; OECD, 2023^[7]). In the case of particularly disruptive developments, like the rise of AI, some central authorities have issued targeted, technology-specific guidance. Singapore and Finland, for example, were among the many education systems that developed national guidance for teachers on the responsible use of AI in classrooms (see Box 2 below).

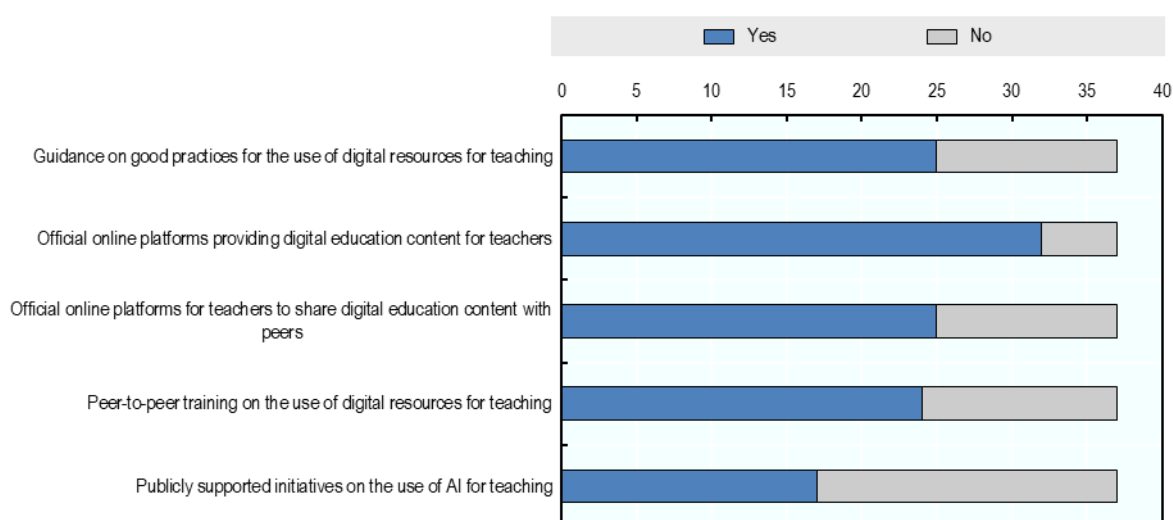
Nearly all education systems provide some form of central support for teachers' effective use of digital resources

Even in education systems with a high degree of decentralisation or pedagogical autonomy, central authorities can use a range of policy levers or targeted initiatives to promote the effective use of digital technologies for teaching and learning. All 37 jurisdictions responding to the Policy Survey reported having some central initiatives in place to support teachers' use of digital resources (Figure 3). These initiatives commonly involve platforms for instructional materials and professional learning resources. In 32 jurisdictions, teachers have access to official online platforms providing access to digital education content and 25 jurisdictions have platforms that allow teachers to share digital education content with their peers. Examples of such platforms can be found in France and Wales (United Kingdom):

- In **France**, the Education Ministry's Digital Education Department operates an array of online platforms and networks for teachers, including *Édubase* and the *Réseau des Concepteurs*. *Édubase* is a national bank of digital educational resources designed for teachers, trainers and inspectors in primary and secondary education. It provides access to pedagogical guides developed by teachers and validated by inspectors for all subjects (Ministère de l'Éducation nationale, 2024^[26]; Ministère de l'Éducation nationale, 2025^[27]). The *Réseau des Concepteurs* is a national network for the co-design and curation of digital learning scenarios (Ministère de l'Éducation nationale, 2025^[28]; Académie de Normandie, 2025^[29]). The content produced by the network is accessible through the online platforms *Magistère* (for trainers) and *Éléa* (for teachers) (Ministère de l'Éducation nationale, 2025^[30]; Ministère de l'Éducation Nationale, 2025^[31]). The network ensures that resources made available via *Magistère* and *Éléa* are pedagogically robust, practically relevant and aligned with national curriculum goals.
- In **Wales (United Kingdom)** the national teaching and learning platform *Hwb* provides educators with access to bilingual resources, opportunities for collaboration and Microsoft Office software (Welsh Government, 2025^[32]). During the COVID-19 pandemic, *Hwb* played an important role in ensuring the continuity of education by providing free access to its virtual learning environment. Although engagement increased substantially during this time, an evaluation suggested that schools with higher proportions of students eligible for free school meals showed lower levels of participation (Sandu and Hampton, 2022^[33]).

Figure 3. Initiatives to foster the effective use of digital resources for teaching

Number of jurisdictions that reported having the following initiatives in place at the central level



Note: Number of jurisdictions with available data = 37. See Annex Table B.1 for further details and individual jurisdictions' responses corresponding to this figure. The data reflect the digital education policies of participating jurisdictions in January 2025. More recent initiatives are therefore not reflected.

Source: Reproduced from Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Figure 2.1.

In addition to online platforms, 25 of 37 education systems participating in the Policy Survey reported that they provide teachers with guidance on good practices for the use of digital resources and 24 systems reported supporting peer-to-peer training on the subject. 17 systems have also supported initiatives on the

use of AI for teaching (Figure 3). In practice, these approaches can complement each other. For example, high-performing systems such as Estonia, Finland and Singapore have established overarching initiatives on AI that encompass guidelines, support platforms and training to ensure that teachers are prepared to deal with its consequences and to shape its effective use in the classroom (see Box 2).

Box 2. Central initiatives to support teachers' effective use of AI

Estonia's "AI Leap 2025" programme

With its "AI Leap 2025" programme, Estonia plans to provide both students and teachers with access to AI-based education tools and to teach them how to use them effectively. The programme's pilot phase will start in September 2025 under the leadership of the Ministry of Education and Research. It will be implemented by a foundation co-founded by the government and private sector actors. Initially, a cohort of 20 000 high school students (Years 10 and 11) and 3 000 teachers will get access to a selection of AI-powered learning applications and receive training on their effective use (e-Estonia, 2025^[34]; Eurydice, 2025^[35]). In 2026, the initiative will expand to include an additional 38 000 students and 2 000 teachers, including those in vocational education. The goal of the initiative is to make teaching more efficient, personalised, diverse and inclusive (e-Estonia, 2025^[34]; TI-Hüpe, 2025^[36]).

The Finnish National Agency for Education's support for teachers' AI skills

The Finnish National Agency for Education (OPH) has set up an online hub ("Support for the Development of AI Skills") to share resources from a range of sources to support teachers in making effective use of AI (OPH, 2025^[37]). This includes recommendations developed by OPH and the Ministry of Education and Culture on "Artificial Intelligence in Early Childhood Education and Care", which provide educators with an overview of their legal obligations concerning AI and general guidance on its effective use (OPH, 2025^[38]). The site also provides access to relevant materials developed by third parties, such as the "AI Guide for Teachers" developed by *Faktabaari*, a fact-checking NGO (Aslama Horowitz et al., 2025^[39]).

Singapore's Generative AI Guides and the AI in Education Ethics Framework

Singapore's EdTech Masterplan 2030 outlines how schools can leverage technology to enhance teaching and learning. One of its five "strategic thrusts" is to provide teachers with professional development opportunities on the effective use of EdTech and to strengthen the EdTech expertise of senior education staff (Curriculum Specialists, Master Teachers [MTT] and Lead Teachers [LT]) (Singapore Ministry of Education, 2025^[40]). To support its implementation, Singapore's Ministry of Education has provided teachers with a suite of AI learning tools through the Student Learning Space (SLS), the MOE's central digital resource hub. These tools are complemented by a series of Generative AI Guides on the safe and effective use of generative AI for teachers, as well as the "AI in Education Ethics Framework", which guides teachers to use AI-tools in line with the principles of agency, inclusivity, fairness and safety (SLS, 2025^[41]; SLS, 2025^[42]).

Sources: TI-HÜPE (TI-Hüpe, 2025^[36]), *AI Leap*, <https://www.aileap.ee/> (accessed on 30 June 2025); OPH (2025^[37]), *Tukea tekoälyosaamisen kehittämiseen* [Support for the Development of AI Skills], <https://www.oph.fi/fi/tukea-tekoalyosaamisen-kehittamiseen> (accessed on 30 June 2025); Singapore Student Learning Space (2025^[43]), *About AI in SLS*, <https://www.learning.moe.edu.sg/ai-in-sls/about-ai-in-sls/> (accessed on 30 June 2025).

In some education systems, efforts to support teachers in the digital transition are developed and co-ordinated by central agencies specifically dedicated to digital education (OECD, 2023^[21]). Examples of such agencies include the following:

- **Spain's** National Institute of Educational Technologies and Teacher Training (*Instituto Nacional de Tecnologías Educativas y de Formación del Profesorado*, INTEF) is responsible for promoting the use of digital education technologies, co-ordinating professional collaboration and innovation among teachers, as well as developing and providing open educational resources and professional learning activities. For example, INTEF offers training opportunities on digital skills, guidelines on digital competency and other digital-related initiatives to a variety of stakeholders, including teachers, school leaders, students and parents (Ministerio de Educación, 2025^[44]).
- In the **Netherlands**, Kennisnet is a central public agency tasked with supporting the effective use of digital technology in the school sector. Kennisnet is funded by the Dutch Ministry of Education, Culture and Science (OCW) with around EUR 13.5 million per year. Its approach is based on the “Four in Balance” model, which emphasises balancing ICT infrastructure, digital learning materials, vision and expertise to align a school's approach to digital education with its overall pedagogical vision (Schouwenburg, 2022^[45]). Kennisnet offers an array of services for teachers, including professional development opportunities and a free platform to create, share and download digital education materials (*Wikiwijs*). Kennisnet also facilitates networks of professionals involved in the implementation of education technologies (i-coaches and ICT co-ordinators) and supports efforts to monitor and evaluate the use of ICT in schools (Kennisnet, 2025^[46]).
- *Oide* is **Ireland's** comprehensive professional development service for teachers. *Oide's* Technology in Education Division (*Oide* TiE) supports teachers' use of digital resources through online platforms with subject-specific educational contents, peer-learning and training opportunities as well as guidance on best practices in the use of digital technology (Department of Education, 2025^[47]; *Oide* Technology in Education, 2025^[48]).
- In the **Flemish Community of Belgium**, the Digisprong Knowledge Centre of the Department of Education and Training serves as a central hub for digital education. It provides expertise, guidance and resources for school leaders, teachers and technical support staff. This includes the KlasCement platform for open educational resources (Digisprong Knowledge Centre, 2025^[49]), as well opportunities for peer learning and professional co-operation (Department of Education and Training, 2025^[50]).

4. Updating professional standards and appraisal frameworks to reflect the role of digital resources in teachers' work

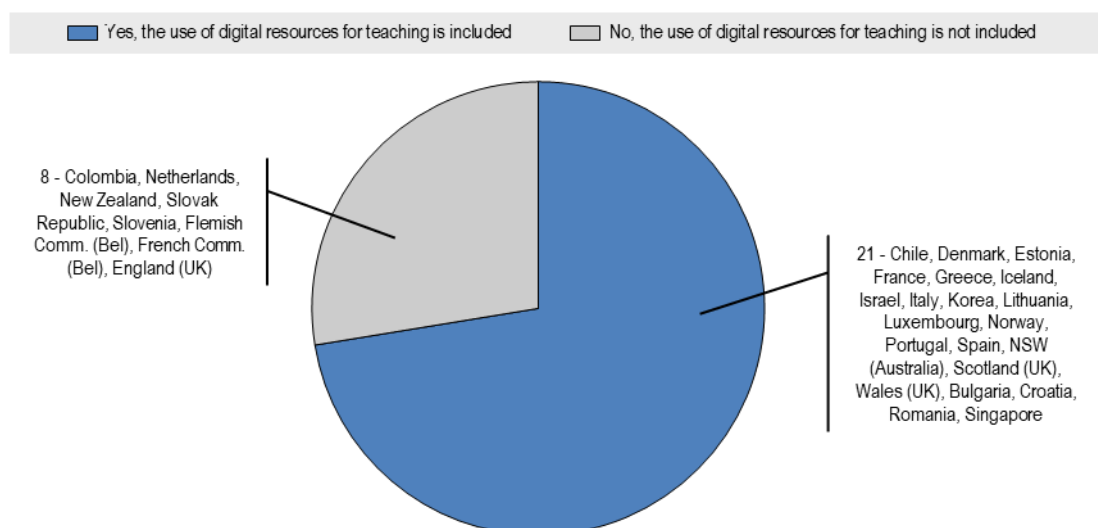
Professional standards and appraisal frameworks are used in many education systems to set expectations for high-quality teaching and to specify the competencies that successful teachers need (Révai, 2018^[51]; OECD, 2018^[52]; OECD, 2019^[53]). They can be effective tools to integrate different elements of the teaching career, from initial teacher preparation and continuing professional learning to teachers' appraisal and career progression (OECD, 2025^[3]). Professional standards can also guide teachers in their self-evaluation and independent learning activities (OECD, 2019^[54]). Updating these standards to reflect the increasing presence of digital resources in schools is important to ensure that the profession remains future-ready and to encourage teachers' effective use of digital resources (Révai, 2018^[51]; OECD, 2018^[52]; OECD, 2019^[53]).

Most education systems refer to the use of digital resources in teachers' professional standards

Most jurisdictions participating in the Policy Survey (29 of 37) have professional standards or formal descriptions of teachers' roles and responsibilities at the central level. Of these 29 systems, 21 include the use of digital resources for teaching in their professional standards, compared to only 8 systems that do not (Figure 4). Several high-performing education systems, including Singapore and Estonia, not only incorporate digital skills in their teachers' general professional standards but complement them with additional frameworks providing more detailed expectations for teachers' effective use of digital resources (see Box 3).

Figure 4. The use of digital resources in teachers' professional standards or role descriptions

Number of jurisdictions that include the use of digital resources in teachers' professional standards or their formal role descriptions (ISCED 2)



Note: Number of jurisdictions with available data = 29. See Annex Table B.2 for further details and individual jurisdictions' responses corresponding to this figure. There were no differences between lower-secondary (ISCED 2), primary (ISCED 1) and upper-secondary education (ISCED 3).

Source: Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.1.

Box 3. Digital education in teachers' professional standards and competency frameworks

Estonia's Digital Competence Framework for Teachers

The initial education, induction and continuing professional learning of Estonian teachers is guided by Professional Qualification Standards, which outline the skills, knowledge and attitudes that teachers need to be successful. Besides six compulsory competencies, the framework includes the "implementation of digital education" as one of two optional domains (Eurydice, 2025^[55]). To further operationalise these expectations, Estonia has adopted a Digital Competence Framework that provides targets for both teachers and students. The framework is aligned with the European Digital Competence Framework for Educators (DigCompEdu) and covers six dimensions for teachers:

1. Professional development and engagement (i.e. communication, co-operation, reflection and professional development using digital technologies); 2. Digital resources (i.e. choosing, creating and sharing digital learning materials); 3. Teaching and learning; 4. Assessment; 5. Empowering learners (i.e. using digital technologies to actively engage learners, to support differentiation, individualisation, and the development of learners' general competences and skills); and 6. Facilitating learners' digital competence (Estonia Education, 2025^[56]).

The Digital Competence Framework is used by the National Education and Youth Board of Estonia (Harno) when designing continuing professional development programmes as well as diagnostic online assessment instruments for teachers. The framework's development is co-ordinated by a Digital Competence task force and Harno as well as an advisory board that includes experts from leading universities and schools. The group meets monthly to adapt, validate and pilot elements of the Digital Competence Framework and associated assessment instruments (Estonia Education, 2025^[56]).

The Singapore Teaching Practice framework and the SkillsFuture for Educators initiative

The Singapore Teaching Practice (STP) is an overarching framework that articulates educators' core beliefs about teaching, learning and student-centric education. It covers Pedagogical Practices (PP), Knowledge Bases (KBs) and the Singapore Curriculum Philosophy (SCP). The SCP highlights the importance of creating appropriate learning experiences that allow students to interact and grow well and safely with others in physical and digital spaces. The STP complements policies and programmes focused specifically on future-oriented skills and digital education, such as the SkillsFuture for Educators (SFEd) initiative (Academy of Singapore Teachers, 2025^[57]).

Launched in 2020, the SFEd is designed as a professional development roadmap to guide teachers in strengthening their practice in six priority areas, including e-pedagogy (i.e. "creating new learning possibilities afforded by digital technologies"). The SFEd defines four levels of practice in this area, which allow teachers to benchmark their proficiency in leveraging digital technologies to accelerate and deepen students' learning by making it more active and personalised (Academy of Singapore Teachers, 2025^[58]). Teachers are encouraged to use the SFEd to identify skills that they want to further develop and to create professional development plans to do so.

Sources: Education Estonia (2025^[56]), *Digital competence: Empowering teachers and students*, <https://www.educationestonia.org/innovation/digital-competence/> (accessed on 1 July 2025); Academy of Singapore Teachers (2025^[58]), *SkillsFuture for Educators: Overview*, <https://academyofsingaporeteachers.moe.edu.sg/professional-excellence/skillsfuture-for-educators/overview/> (accessed on 1 July 2025).

To ensure that professional standards remain relevant amid rapid technological change, some education systems have engaged in structured exercises to explore how teachers' roles may evolve in the future (OECD, 2024^[4]). As part of the OECD project "New Professionalism and the Future of Teaching", for

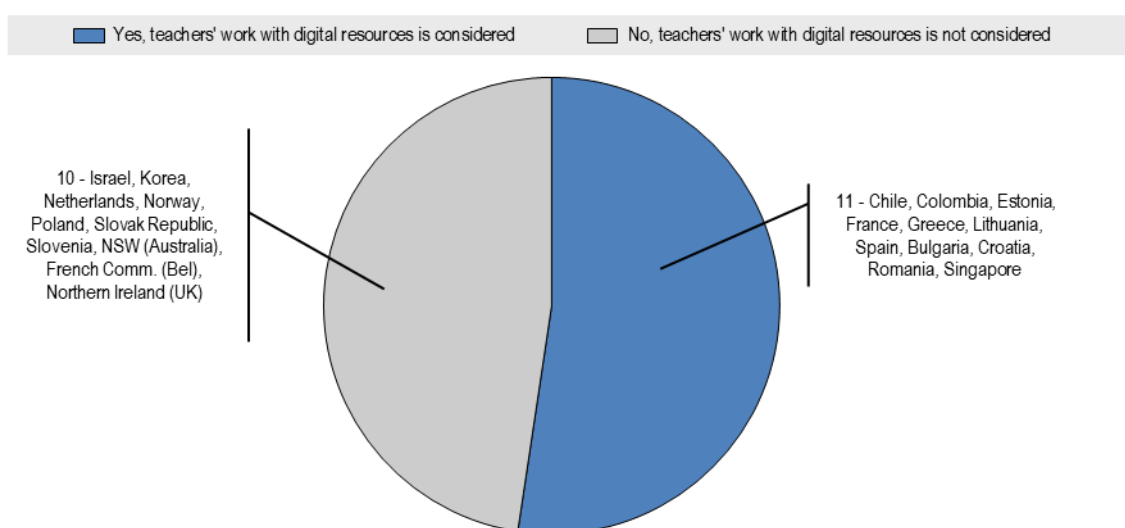
example, the **Flemish Community of Belgium** explored how different “teacher personas” might be affected by scenarios such as the increased use of AI and other digital tools in schools. The Flemish Community also engaged various stakeholders in “ambition loop” exercises to reflect on short-term improvements, the longer-term transformation of the teaching profession and positive scenarios for the future of teaching and schooling (OECD, 2024, p. 29^[59]).

Only around half of all central teacher evaluation and appraisal frameworks consider teachers’ work with digital resources

The impact of professional standards on teachers’ work ultimately depends on whether they are actively used by relevant actors as a reference point in evaluating, developing and improving teachers’ work, for example in the context of their regular appraisal (Boeskens and Nusche, 2021, p. 65^[60]). However, although most education systems have integrated references to digital technologies in teachers’ professional standards, results of the Policy Survey indicate that many do not systematically consider teachers’ use of digital resources when evaluating their work. Of the 36 responding jurisdictions with available data, 21 reported having a central framework for teacher evaluation.² Of these frameworks, only 11 (including those of Estonia and Singapore) considered teachers’ work with digital resources, whereas 10 (including those of Korea and NSW [Australia]) did not (Figure 5).

Figure 5. The role of digital education in teacher appraisal and evaluation frameworks

Number of jurisdictions that consider teachers' work with digital resources in their formal appraisal/evaluation frameworks



Note: Number of jurisdictions with central appraisal/evaluation frameworks = 21. See Annex Table B.2 for further details and individual jurisdictions' responses corresponding to this figure.

Source: Boeskens, L. and K. Meyer (2025^[2]), “Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age”, *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.1.

Evaluating teachers’ digital competency and their use of digital resources as part of their regular appraisal can be a means to provide recognition for successful practices and to support teachers in strengthening their digital skills (OECD, 2023^[21]; OECD, 2019^[54]). Evaluation criteria can also encourage educators to engage in innovative practices and to proactively explore the integration of digital resources, particularly if evaluations are linked to opportunities for professional advancement or other incentives (OECD, 2023^[21];

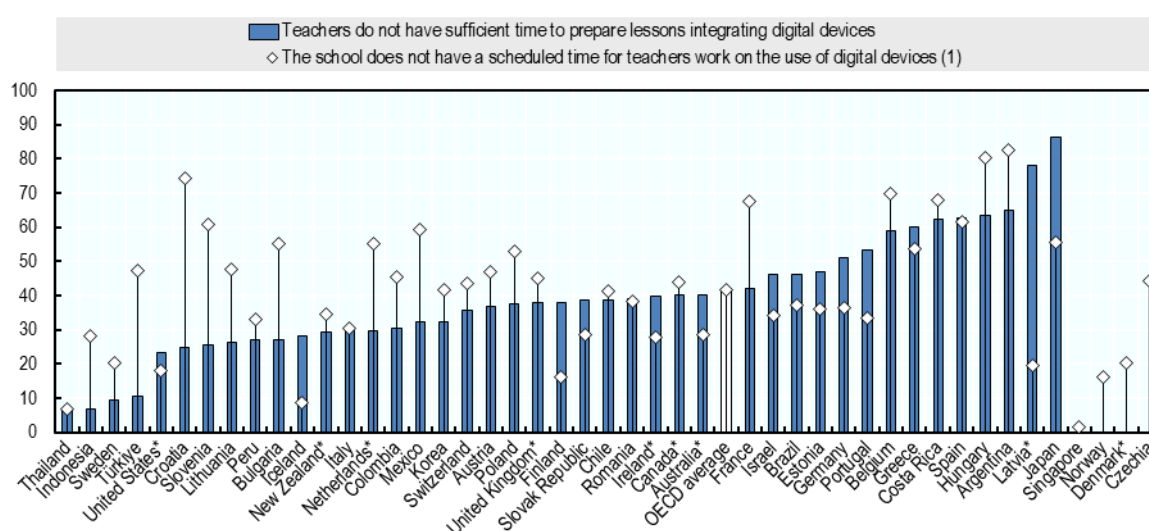
OECD, 2023^[7]). For example, in **Croatia** and **Romania**, teachers' engagement with digital education resources is considered in their evaluation (Ministry of Science and Education, 2019^[61]; Ministerul educației și cercetării, 2020^[62]) and is taken into account for career progression opportunities and awards, like Romania's "Diploma for Innovation and Digitalization in Education" (Ministry of Science and Education, 2023^[63]; Parlamentul României, 2023^[64]).³ Of course, even in the absence of explicit evaluation criteria, principals and other school staff may appraise teachers' ability to use digital resources effectively in the classroom, provided that they are well-prepared to identify and evaluate such practices.

5. Managing teachers' workload and making time for their engagement with digital resources

Teachers require time to learn about and evaluate new digital technologies, to strengthen their digital skills and to adapt their teaching practices to take advantage of the digital transformation (Darling-Hammond, Hyler and Gardner, 2017^[65]; OECD, forthcoming^[10]). In practice, many schools report that teachers lack the time to familiarise themselves with digital tools and to integrate them into their work. In PISA 2022, 41% of 15-year-old students across OECD countries attended schools whose principal reported that teachers did not have sufficient time to prepare lessons integrating digital devices. This proportion varied widely, ranging from less than 10% in Thailand, Indonesia and Sweden to more than 70% in Latvia and Japan (Figure 6).

Figure 6. Teachers' time to work with digital devices for education (2022)

Percentage of 15-year-old students whose school leader reported the following about their school



Note: (1) School leaders were asked whether the school has "a scheduled time for teachers to meet to share, evaluate or develop instructional materials and approaches that employ digital devices".

Source: OECD (2023^[5]), *PISA 2022 Results (Volume II): Learning During – and From – Disruption*, <https://doi.org/10.1787/a97db61c-en>, Tables II.B1.5.32 and II.B1.5.35.

Teachers' reported lack of time underlines the significant investments that teachers need to make to adapt their work to digital resources and points to a challenge that policymakers need to consider. Adapting

teachers' schedules to provide them with more time to engage in professional learning and to collaborate with peers on the use of digital resources is one strategy to address this problem (Darling-Hammond, Hyler and Gardner, 2017^[65]). In PISA 2022, 58% of students attended schools whose principal reported scheduling time for teachers to share, evaluate or develop instructional materials and approaches that employ digital devices (Figure 6). Across countries, however, having a scheduled time for work on digital resources was only weakly correlated with the perception that teachers have sufficient time for this work. Other factors that can help teachers in making time to engage with digital resources are discussed in the following sections.

Adapting teachers' schedules or encouraging schools to make time for work on digital resources

In many OECD countries, local authorities or school leaders have some flexibility in adapting teachers' schedules (Boeskens and Nusche, 2021^[60]). This can allow school leaders to balance local demands and staff needs with central-level goals as they advance the integration of digital resources (OECD, 2023^[7]). At the same time, central authorities can encourage schools to allocate time for selected teachers to advance the implementation of digital resources, to encourage peer learning and to share their knowledge with other staff. Examples of this practice can be seen in the Netherlands and in Finland:

- In the **Netherlands**, schools can designate teachers to serve as ICT co-ordinators or i-coaches and to assume responsibilities in developing their schools' ICT policies and vision for digital education, supporting the professional development of staff and focusing on the pedagogical use of digital resources. *Kennisnet*, the central agency for digital education (see above), facilitates training and professional networks for ICT co-ordinators (*Kennisgroep ICT*) and i-coaches (*Kenniskring i-coaches*) in partnership with the primary and secondary education councils (*PO-Raad* and *VO-raad*). Schools have wide-ranging autonomy over the use of their resources and no earmarked budget is provided for i-coaches or ICT co-ordinators. However, some schools allocate resources or targeted funding to free up teaching time (e.g. an hour per week) for staff with responsibility for ICT to advance their schools' use of digital resources (OECD, forthcoming^[66]).
- In **Finland**, schools are encouraged to appoint experienced teachers as Tutor Teachers who dedicate some of their working time (around 1 to 5 hours per week) to supporting professional learning and pedagogical improvements in their schools. The Tutor Teachers programme is organised by the Finnish National Agency for Education and one of its objectives is to support pedagogical innovation and to promote the digitalisation of teaching. Tutor Teachers also work on the exchange of best practices in local, regional, national and thematic networks, including networks focused on digital education. Between 2016 and 2019, EUR 33.8 million had been granted to support the Tutor Teachers' activities in basic education (OPH, 2025^[67]; OECD, 2024, p. 89^[4]).

In PISA 2022, the highest rate of students whose teachers have a scheduled time to collaborate on their work with digital devices (98.6%) was seen in Singapore, where many school leaders set aside dedicated time for work in Professional Learning Communities (PLCs) (see Box) (Academy of Singapore Teachers, 2025^[68]). While Singapore's PLCs can cover a broad range of issues, many incorporate reflections on the effective use of digital technologies and – in PISA 2022 – 87% of Singapore's students attended schools with specific programmes to promote teachers' collaboration on the use of digital devices (OECD, 2023, pp. 409, Table II.B1.5.35^[5]).

Box 4. Professional Learning Communities (PLCs) in Singapore

Working in Professional Learning Communities (PLCs) plays an important role in the professional growth of Singapore's teachers. Schools operating as PLCs set aside time slots for teachers to organise themselves in smaller Professional Learning Teams (PLTs). Work in PLTs is oriented towards student learning and guided by four "Critical Questions" (What is it we expect students to learn? How will we know when they have learned it? How will we respond when they haven't learned? How will we respond when they already know it?). Teachers are encouraged to adopt a "Critical Inquiry Method" involving four stages: 1. Identifying the gaps in their teaching and learning, 2. Planning an intervention to address it, 3. Implementing the proposed instructional strategy and collecting data to observe its impact on students' learning, and 4. Reflecting and collectively reviewing the intervention's impact with peers (Academy of Singapore Teachers, 2025^[68]). Though PLCs are not focused on digital education or limited to specific types of interventions, they are effective means to stimulate innovative practices involving technological innovations. PLCs are also aligned with Singapore's EdTech Masterplan 2030, which calls on schools to foster collaboration and for teachers to embrace a culture of teamwork, sharing and adapting digital resources, as well as best practices for teaching with technology (Singapore Ministry of Education, 2025^[40]).

Source: Academy of Singapore Teachers (2025^[68]), *Professional Learning Community (PLC): Overview*, <https://academyofsingaporeteachers.moe.edu.sg/professional-excellence/our-plc-journey/overview/> (accessed on 1 August 2025).

Promoting teachers' collaboration on the use of new technologies or giving them time to familiarise themselves with digital pedagogies does not imply promoting digital learning for its own sake. In fact, PISA 2022 data show that the use of these strategies is rarely associated with students spending more time learning with digital resources at school (OECD, forthcoming^[6]). Instead, giving teachers time to engage with digital innovations and to collaboratively explore their effective use might promote a more deliberate, targeted and student-centred integration of digital tools in the classroom.

Facilitating or automating repetitive and administrative tasks

While the digital transformation of schools places new demands on teachers, it also has the potential to reduce teachers' work on repetitive or administrative tasks. This could free up educators' time for pedagogical work with digital education resources, instructional planning or individual student support (Elmoazen et al., 2023^[69]; Martínez-Comesaña et al., 2023^[12]). **Korea**, for example, uses its "National Education Information System" (NEIS) to combine student information, administrative functions and a digital credential system in a centralised web-based platform (Ministry of Education and Human Resources Development, 2025^[70]). The NEIS digitalises and automates administrative tasks to reduce the time teachers spend on data entry, record keeping or verification procedures. Processing data from over 12 000 public and private schools and 17 regional education offices, the system generated annual savings of USD 237 million, based on a 2010 estimate (OECD, 2023, p. 25^[21]). **England (United Kingdom)** has pursued similar goals, using a range of digital tools to improve teachers' workload and well-being (see Box 5).

Box 5. Improving teachers' workload and well-being through digital tools in England (UK)

In 2018, England's Department for Education (DfE) launched the "School Workload Reduction Toolkit", which provided schools with advice and digital tools to streamline a range of processes, including administrative tasks, lesson planning, data management and marking (Department for Education, 2023^[71]). A 2020 evaluation of the toolkit found that its initiatives had a positive impact on teachers' well-being and reduced the time spent on the targeted tasks without weakening students' outcomes (Richard Churches, 2020^[72]). In 2024, the DfE expanded on the toolkit's offer with the "Improve Workload and Wellbeing for School Staff" service, which placed greater emphasis on staff well-being and improved the accessibility of the toolkit's resources. The service seeks to help schools and educators to identify and address workload-related issues and to evaluate the effectiveness of the interventions they implement. It also offers a platform for schools to share successful initiatives and to learn from their peers (Department for Education, 2024^[73]).

To further enhance administrative efficiency and to support school leaders and school business professionals, DfE launched the DfE Connect platform. DfE Connect is designed to streamline access to DfE services and make it easier for school staff to fulfil their administrative responsibilities, to access DfE grant and funding opportunities, to manage school finances and to obtain information on best practices (Department for Education, 2025^[74]).

The DfE also invested in the development of AI tools to assist teachers with tasks such as marking, lesson planning, record keeping and student feedback, in order to increase their time for face-to-face teaching. For example, in 2023, GBP 2 million were invested to support Oak National Academy in developing "Aila", an AI-powered lesson assistant that helps teachers to quickly create personalised lesson plans (Oak National Academy, 2024^[75]; Department for Education, 2023^[76]). In 2025, the UK government announced an additional GBP 1 million investment in 16 EdTech companies to support the development of AI-powered assessment tools designed to assist teachers with grading and providing student-specific feedback (Department for Education, 2025^[77]).

Source: Department for Education (2024^[73]), *Improve workload and wellbeing for school staff*, <https://improve-workload-and-wellbeing-for-school-staff.education.gov.uk/> (accessed on 19 February 2025).

6. Adapting career structures and allocating responsibilities to manage the use of digital resources in schools

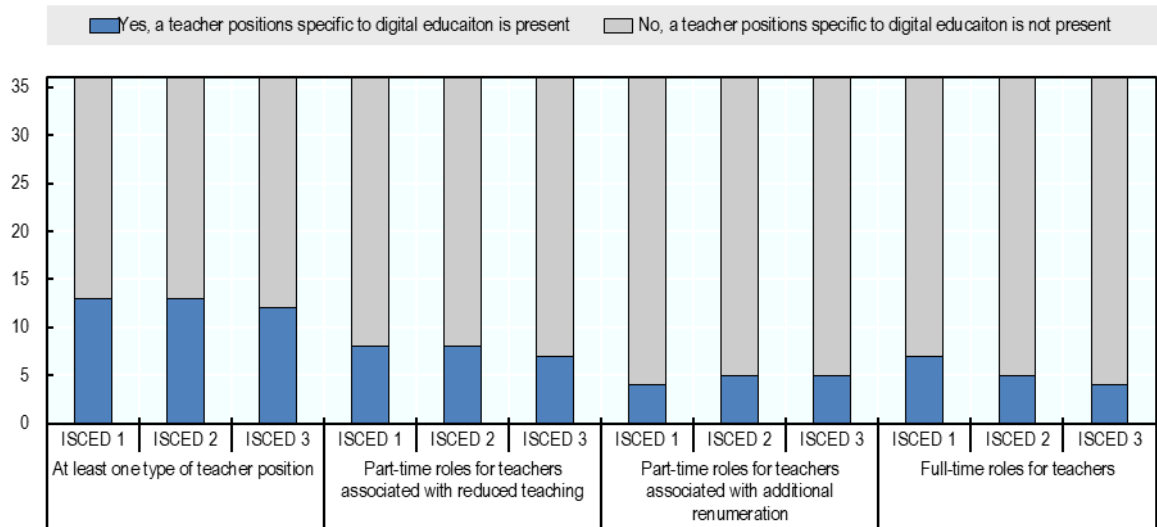
Over the past decades, an increasing number of education systems have created formal staff positions in schools (roles such as “digital champion” or “IT co-ordinator”) to foster innovation and advance the integration of digital technologies into teachers’ work (OECD, 2024, p. 88^[4]). This trend has allowed schools to take a more strategic approach to digital education while creating opportunities to recognise and reward teachers’ effective use of educational technologies and to support their professional growth (OECD, 2023^[21]; OECD, 2023^[7]). Roles with a focus on digital education can be integrated into teachers’ careers vertically (i.e. as successive career stages with a progressive increase in responsibilities) with positions such as digital mentor or deputy school leader; or horizontally (i.e. through functional or thematic specialisation) with positions such as digital co-ordinator (OECD, 2019^[54]; OECD, 2023^[7]).

Some systems assign responsibilities for the use of digital education resources to teachers, while others turn to technical support staff

Co-ordinating the use of digital education resources in schools requires a combination of both technical and pedagogical expertise (European Commission, 2019^[78]; OECD, forthcoming^[10]). This is reflected in the different staff profiles that school systems entrust with responsibilities related to the use of digital resources for teaching. Results of the Policy Survey show that a sizeable proportion of systems (13 of 36 jurisdictions) assign these positions to teachers – usually to ICT teachers or staff with a particular interest in digital education (see Box 6 for an example from Singapore) (OECD, 2023^[7]). In 5 systems, these positions are full-time roles but more frequently (in 11 systems), teachers assume these positions part-time (at ISCED 2). Among the 11 jurisdictions with part-time positions, 8 compensate teachers for their additional work with a reduction in teaching hours, while 5 provide teachers with additional remuneration (at ISCED 2) (Figure 7).

Figure 7. Dedicated teacher positions related to the use of digital resources

Number of jurisdictions that reported having dedicated staff positions for teachers with responsibilities related to the use of digital resources for teaching

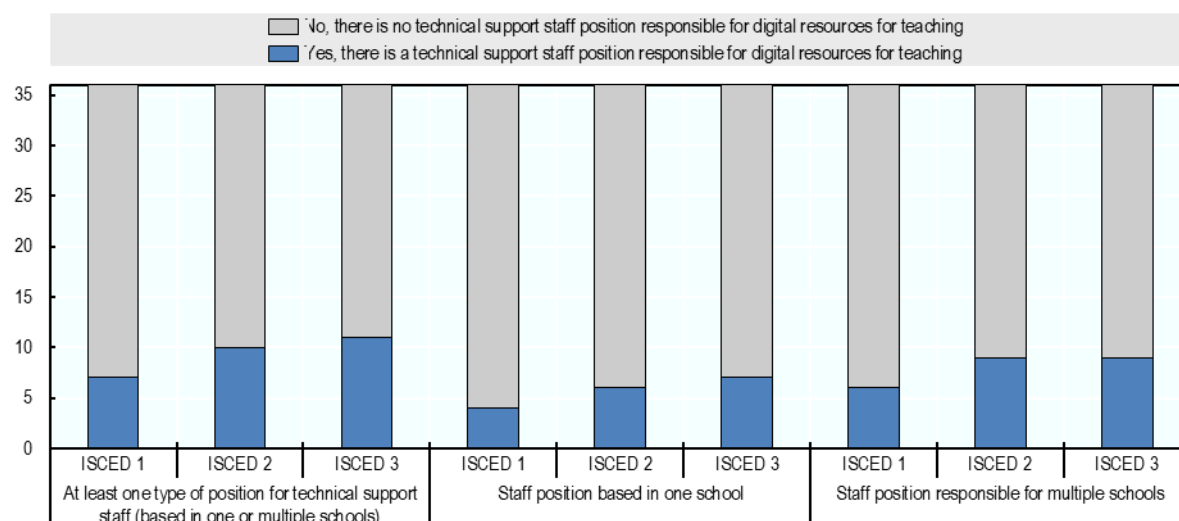


Note: Number of jurisdictions with available data = 36. See Annex Table B.3 for further details and individual jurisdictions' responses corresponding to this figure.
Source: Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.2.

A similar number of education systems (10 of 36 at ISCED 2) assigns responsibilities for the use of digital resources for teaching to technical support staff, rather than (or in addition to) teachers (Figure 8). This choice could reflect a greater emphasis on technical rather than pedagogical aspects of teaching with digital resources or efforts to complement teachers' expertise with external technical capacity. In 6 systems, these positions are assigned to a single school and in 9 systems, they are assigned to multiple schools. Several systems rely on a combination of school-based positions and staff responsible for multiple schools, depending on the context. In some cases, for example in the Flemish Community of Belgium, New Zealand and Singapore, more than one of these models coexist (see Annex Table B.3 for a detailed overview).

Figure 8. Technical support staff positions for the use of digital education resources

Number of jurisdictions that reported having dedicated positions for technical support staff with responsibilities related to the use of digital resources for teaching



Note: Number of jurisdictions with available data = 36. See Annex Table B.3 for further details and individual jurisdictions' responses corresponding to this figure.

Source: Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.2.

Creating staff positions dedicated to the use of digital resources is often at the discretion of schools or local authorities

In many of the jurisdictions participating in the Policy Survey (26 of 36), schools or local authorities have some discretion in creating staff positions with responsibilities for the use of digital resources. Furthermore, in 7 jurisdictions (at ISCED 2) the presence of such roles depends on school characteristics (e.g. their size or educational offer). Many systems also use different approaches, depending on the roles in question, or define staff positions at the central level but leave it to schools to decide whether to fill them (Boeskens and Meyer, 2025, pp. 35, Figure 7.2^[2]). Examples of such staff positions, in addition to those in the **Netherlands** and **Finland** (discussed above), include the following:

- In 2015, **Italy's** National Digital School Plan (PNSD) established a role for digital animators (*animatori digitali*) in each school. The animators are designated teachers who receive training to work with their school management team to promote digital innovations aligned with the PNSD. Schools receive an annual budget of EUR 1 000 to fund activities led by digital animators, which can include internal trainings, workshops involving the wider school community, or the implementation of innovative digital solutions (Ministero dell'Istruzione e del Merito, 2015^[79]; Ministero dell'Istruzione e del Merito, 2025^[80]). Over 8 200 teachers now serve as digital animators (Ministero dell'Istruzione e del Merito, 2025^[81]). In a 2022 evaluation of the PNSD, 95% of the surveyed digital animators reported feeling that their work increased teachers' digital skills and 26% felt that their work had significantly increased the adoption of innovative teaching methodologies (Agasisti et al., 2023^[82]).
- In 2022, the **Slovak Republic** created the role of digital co-ordinators (defined in the Slovak Act no. 138/2019), who are employed in schools to provide teachers with support, guidance and training on digital education (UNESCO, 2025^[83]). Digital co-ordinators also create, curate and

share digital education materials, manage digital resources and ensure cybersecurity in their schools (Ministersrva Spravodlivosti Slovenskej Republiky, 2024^[84]). Digital co-ordinator positions can be funded with schools' own resources or with central funding from a national project supporting various teaching and non-teaching positions in schools.

- In the **French Community of Belgium**, teachers can assume the role of digital delegate (*délégué-référent au numérique*, DRN) to promote the effective use of digital resources in their schools. DRN receive targeted training and school providers (*pouvoirs organisateurs*) receive central resources to reduce the DRNs' teaching time (Fédération Wallonie-Bruxelles, 2019^[85]; Gouvernement de la Communauté française, 2019^[86]; Wallonie-Bruxelles Enseignement, 2021^[87]). In 2021/22, 30% of primary schools decided to provide DRNs with reduced teaching hours (2.6 periods on average), as did 64% of secondary schools (7.1 periods on average) (Cour des Comptes, 2024^[88]).

Team-based approaches and co-operation between staff at different levels of the system can strengthen the support for digital education in schools

In practice, the approaches described above are not mutually exclusive and responsibilities for digital resources in schools can be shared between different types of staff. In the Policy Survey, several education systems (5 in primary education and 9 in secondary education) reported that both teachers and technical support staff can assume responsibilities for the use of digital education resources for teaching (see Annex Table B.3). Staff with technical and pedagogical skills can complement each other in strengthening the effective use of digital education in schools. **Singapore** provides an example of this approach, demonstrating how staff at multiple levels of the education system can work together to support digital education with their respective expertise (see Box 6).

Box 6. Co-ordinating support for digital education across multiple levels in Singapore

In Singapore, support for digital education is co-ordinated across multiple levels of the education system. At the central level, two divisions within the Ministry of Education (MOE) are dedicated to supporting the use of ICT in schools. The Educational Technology Division (ETD) is responsible for providing thought leadership on the effective use of educational technology, for designing and implementing user-centric applications, for building the digital capacity of school staff and for developing digital content to support the curriculum, assessment and professional development (Singapore Ministry of Education, 2025^[89]). The MOE's Digital Excellence & Products Division (DXD) plays a complementary role, overseeing the deployment of IT services, systems and infrastructure to support administration, management and education. Its responsibilities include the provision of expert advice on the deployment of information technologies in both the ministry and in schools (Singapore Ministry of Education, 2025^[90]).

All schools are encouraged to appoint a Head of Department (HOD [ICT]) and a school-based ICT Manager/Associate (ICT M/A) at their own discretion and based on their needs. In 2024/25, about 60% of schools had appointed a HOD (ICT) or equivalent and 66% of schools had at least one ICT M/A. The role of the HOD (ICT) is to lead the use of digital resources in teaching and learning at their school and to strengthen the digital competency of teachers. The ICT M/A assume responsibility for the technical co-ordination of their schools' digital systems and for safely and efficiently implementing digital applications and smart school systems. They also support the HOD (ICT) in training school staff on the use of ICT and assisting them in identifying new ICT resources to support teaching and learning (Ministry of Education, 2024^[91]).

The school-based staff are supported by ICT Partners based in the MOE Headquarters, who serve as points of contact between schools and the ministry. ICT Partners communicate with schools, advise them on central-level ICT and security policies, provide technical advice on school-initiated ICT projects and support the development of school-based ICT M/A staff. Each ICT partner is deployed to serve one or two clusters of schools.

Sources: Singapore Ministry of Education (2022^[92]), *Educational Technology Division (ETD)*, <https://www.moe.gov.sg/about-us/organisation-structure/etd> (accessed on 1 August 2025); Singapore Ministry of Education (2025^[90]), *Digital Excellence & Products Division (DXD)*, <https://www.moe.gov.sg/about-us/organisation-structure/dxd> (accessed on 1 August 2025).

Multiple OECD countries have embraced collaborative and team-based approaches to supporting the use of ICT in schools. Such collaborative models can help to distribute technical, pedagogical and administrative tasks related to the use of digital resources and to attract and develop professionals with complementary, specialised profiles (European Commission, 2024^[93]). Collaboration can also spread the associated workload more evenly and avoid placing excessive responsibilities on single members of staff or teachers with part-time roles. Allocating responsibilities for the use of digital resources in schools to a team can also foster professional exchange and opportunities for peer learning between professionals with different sets of skills. In pursuit of these goals, the **Flemish Community of Belgium** has recently started promoting a more team-based approach to ICT in schools (see Box 7).

Box 7. Promoting team-based approaches to ICT in the Flemish Community of Belgium

The Flemish Community of Belgium introduced the role of school-based ICT co-ordinators in 2003 and provided all primary and secondary schools with resources to appoint staff members to the role. Since the COVID-19 pandemic, the Flemish Community sought to further strengthen the effectiveness and capacity of ICT co-ordinators, leading to a European Commission Technical Support Instrument (TSI) project to promote team-based approaches to ICT support in schools (Flemish Parliament, 2023^[94]). The project – implemented with the Flemish Department for Education and Training's Digisprong initiative – recognised that some Flemish schools had already implemented more distributed models where technical, pedagogical and administrative ICT tasks were shared between multiple staff. ICT co-ordinators in these schools were supported by colleagues taking on such tasks as learning environment management, social media co-ordination or media coaching. These team-based models distributed the workload more effectively and were considered to be more aligned with modern forms of distributive, team-based school leadership (European Commission, 2024^[93]).

To facilitate the wider adoption of such team-based approaches and to provide greater coherence to the role of ICT co-ordinators, the project developed a task profile for ICT co-ordinators that clearly defines their roles and responsibilities, led a study examining the characteristics of effective ICT teams, and developed guidelines for a more team-oriented ICT co-ordination in schools (European Commission, 2024^[93]). The study identified that strong ICT teams tended to comprise managerial roles (dedicated to co-ordinating the team as well as elaborating and implementing ICT policy plans), pedagogical roles (focused on the didactic aspects of integrating ICT in the classroom) and technological roles (working on the quality insurance and maintenance of digital resources). The study also found strong ICT teams to be supported by administrative staff, to involve ICT-savvy teachers and to co-operate with students, parents and external partners on ICT in schools (Desmedt, Sempels and Pauwels, 2024, p. 23 f.^[95]).

Sources: European Commission (2024^[93]), *Digital transformation in the Flemish education system: Reforming ICT teams at school*, https://reform-support.ec.europa.eu/what-we-do/skills-education-and-training/digital-transformation-flemish-education-system-reforming-ict-teams-school_en; Flemish Parliament (2023^[94]), *Verslag van de hoorzitting over ICT-coördinatie in het onderwijs* [Report on the Hearing on ICT Co-ordination in Education], [Verslag van de hoorzitting 1896 \(2023-2024\) nr.1 | Vlaams Parlement](#) (accessed on 28 February 2025).

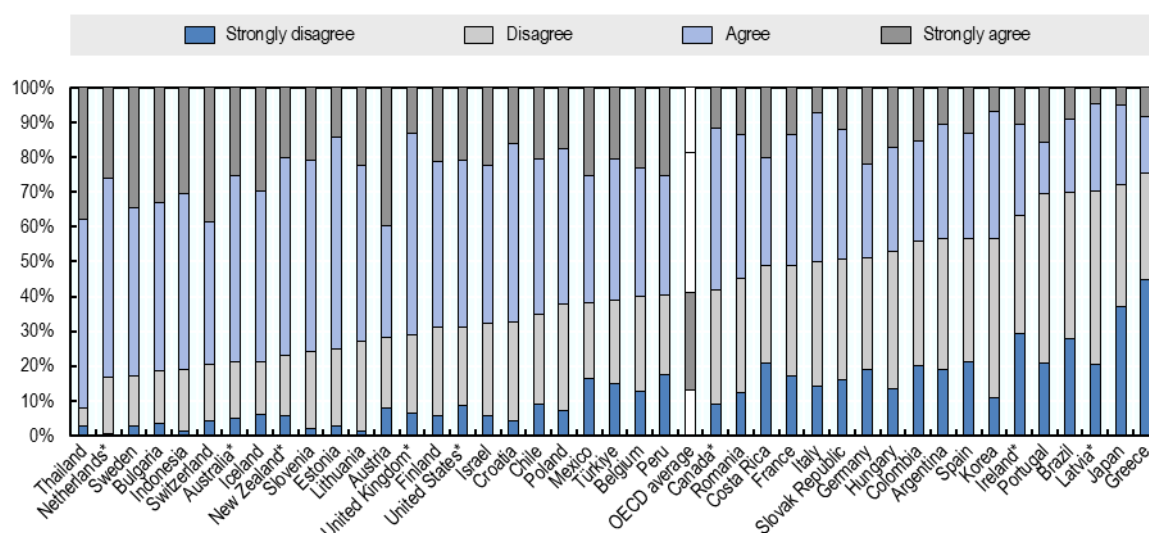
7. Providing technical assistance to schools and ensuring the maintenance of digital resources

Schools' access to technical assistant staff remains a challenge in many OECD countries

Providing teachers and students with a reliable digital infrastructure is not a one-time effort but requires the continuous maintenance of digital equipment and software (OECD, 2023, p. 145^[7]). School leaders in many OECD countries express concerns about a lack of technical support in addressing this challenge. In PISA 2022, 41% of 15-year-old students attended schools whose principals did not think their school had sufficient qualified technical assistant staff, on average across OECD countries, with rates ranging from less than 20% in the Netherlands and Sweden to more than 70% in Greece, Japan and Latvia (Figure 9) (OECD, 2023^[5]). Finding effective solutions for the maintenance of digital resources not only helps to ensure their safe and effective operation but also enables teachers to focus on the pedagogical aspects of their use (European Commission, 2019^[78]).

Figure 9. Qualified technical assistant staff in schools (2022)

Percentage of 15-year-old students whose principal agreed or disagreed that the school has sufficient qualified technical assistant staff



Note: Countries are ordered in ascending order of the share of students whose principals disagreed or strongly disagreed that the school has sufficient qualified technical assistant staff.

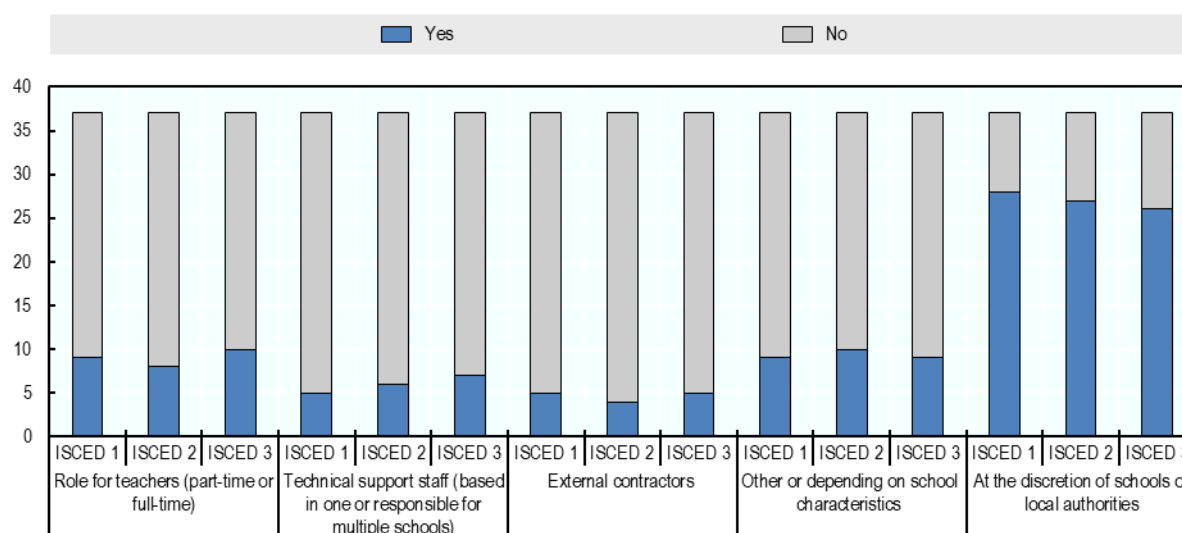
Source: OECD (2023^[5]), *PISA 2022 Results (Volume II): Learning During – and From – Disruption*, <https://doi.org/10.1787/a97db61c-en>, Table II.B1.5.29.

The maintenance of digital resources is usually organised at the school or local levels, but in a third of countries, the responsibility is assumed by teachers

As is the case for the pedagogical integration of digital resources (discussed above), responses to the Policy Survey show that responsibilities for their maintenance are usually assigned at the discretion of schools or local authorities. This is the case in 27 of 37 systems (at ISCED 2), including Estonia, Finland and Singapore (Figure 10). Where maintenance responsibilities are allocated centrally, 8 systems assign them to teachers with a part-time or full-time role and 6 systems assign them to technical staff (at ISCED 2). Among the latter, technical maintenance staff is either based in one school (4 systems) or responsible for multiple schools (5 systems). Fewer systems (4 of 37 at ISCED 2) rely on external contractors to maintain digital resources in schools (Figure 10). These different approaches are not mutually exclusive and some education systems rely one more than one type of staff to assume responsibilities for maintenance duties (see Annex Table B.3 for a detailed overview).

Figure 10. Staff responsible for maintaining digital resources in schools

Number of jurisdictions assigning responsibility for the maintenance of digital resources to different types of staff



Note: Number of jurisdictions with available data = 37. See Annex Table B.3 for further details and individual jurisdictions' responses corresponding to this figure.

Source: Reproduced from Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Figure 7.3.

8. Conclusions

Education authorities have recognised that digitalisation will have important implications for their ability to attract, retain and develop teachers in the coming years. As technology transforms teaching and learning in schools, teacher policies may need to be adapted to provide continued support to the profession and to help educators address new challenges they face in the classroom. Results of the Policy Survey on School Education in the Digital Age indicate that many OECD education systems have already introduced reforms and initiatives aimed at supporting innovative practices and the effective and deliberate use of digital technologies in schools. For example, nearly all responding jurisdictions provide some form of central guidance and support to promote teachers' effective use of digital resources. Such initiatives frequently involve the development of online platforms that allow teachers to access guidance on the effective use of digital resources or to share digital education content and good practices with their peers. Around half of the participating jurisdictions have also launched initiatives specifically focused on the use of AI for teaching.

Professional standards and appraisal frameworks can be another powerful tool to set expectations for high-quality teaching and to guide educators to attain the skills they need to be successful in the digital age. Most jurisdictions participating in the Policy Survey now address the use of digital resources in teachers' professional standards. To ensure that these standards evolve and remain relevant amid rapid technological change, some education systems have engaged in structured reflections with stakeholders to explore how teachers' roles may evolve in the future. By contrast, only half of the participating education systems systematically consider teachers' work with digital resources as part of their regular appraisal.

Digital tools have the potential to make teachers' work more efficient but also require them to spend time to learn about the use of new digital technologies and to adapt their teaching practices accordingly. While setting aside time for teachers to work with digital resources can help them to make informed decisions on their integration in the classroom, it is also important to consider how digital technologies can reduce the time teachers spend on repetitive or administrative tasks. Some countries have used this strategy successfully to ease teachers' workload and to help them focus on improving students' learning outcomes.

Over the past decades, many school systems have created formal roles to foster innovation in schools and to support the effective use of digital technologies. This has allowed schools to take a more strategic approach to digital education and created new opportunities for teachers' professional growth. Schools or local authorities tend to be in charge of filling these positions and allocating responsibilities for both the technical and the pedagogical aspects of digital education. Yet, central authorities play an important role in defining these roles and their responsibilities. Given the complex combination of skills and expertise required to successfully integrate digital resources in schools, several countries have addressed this challenge by promoting promising team-based approaches or by fostering co-operation between staff at different levels of the system.

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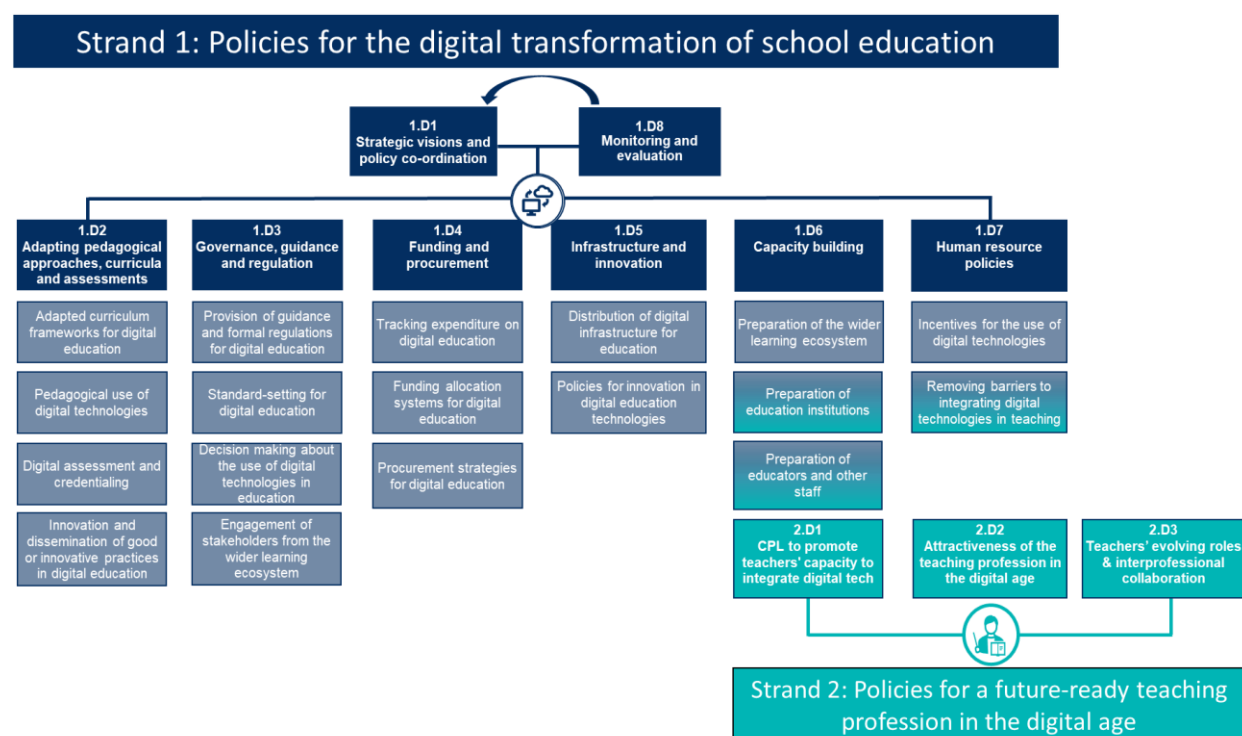
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Annex A. Analytical framework of the Resourcing School Education project

The OECD project “Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers” is guided by an analytical framework (see Figure A.1), which covers its Strand 1 (Policies for the Digital Transformation of School Education) and Strand 2 (Policies for a Future-Ready Teaching Profession in the Digital Age). This policy paper addresses topics that cut across Dimension 1.D1 (“Strategic visions and policy co-ordination”), the Cells “Provision of guidance and formal regulations for digital education” and “Standard-setting for digital education” of Dimension 1.D3 (“Governance, guidance and regulation”), Dimension 1.D7 (“Human resource policies”), as well as the framework’s Dimension 2.D3 (“Teachers’ evolving roles & interprofessional collaboration”).

Figure A.1. Analytical framework guiding the Resourcing School Education project



Note: For a more detailed explanation of the project's analytical framework, see OECD (OECD, 2023, pp. 22, Figure 1.2[7]).

Source: OECD (2025^[11]), *Resourcing School Education: Policies for the Digital Transformation of Education and Future-Readiness of Teachers* [website], <https://www.oecd.org/en/about/projects/resourcing-school-education.html> (accessed on 12 March 2025).

Annex B. Responses to the Policy Survey

This annex includes comparative tables presenting the data from the Policy Survey on School Education in the Digital Age referred to in this policy paper, including the individual responses of all 37 jurisdictions participating in the survey. The tables include detailed information (and contextual footnotes) for all question items covered in the figures and text of this policy paper. For the complete results of the Policy Survey, please refer to Boeskens and Meyer (2025^[2]). The results reflect the digital education policies of participating jurisdictions in January 2025. As a result, some recent policy developments are not reflected in this data.

The following symbols and abbreviations are used in the tables:

- a = Data are not applicable because the category or question does not apply.
- m = Data are missing because no response was provided.

Where footnotes are included on specific jurisdictions' responses, the cells that they refer to are marked with an asterisk (*)

Annex Table B.1. Policies on the pedagogical use of digital resources

Inclusion of digital education in school-level strategies and initiatives to foster the effective use of digital resources for teaching

	Requirements for schools to develop a plan or strategy concerning the use of digital resources for teaching (1)	Central initiatives to foster the effective use of digital resources for teaching					
		Official guidance on good practices for the use of digital resources for teaching	Official online platforms providing selected digital education content for teachers (2)	Official online platforms for teachers to share digital education content with peers	Publicly supported peer-to-peer training on the use of digital resources for teaching	Publicly supported initiatives on the use of AI for teaching	Other
OECD countries							
Austria	Yes (B)		✓	✓	✓	✓	
Chile	Yes (B)	✓	✓	✓	✓	✓	
Colombia	No	✓	✓	✓	✓		
Denmark	No	✓	✓				
Estonia	Yes (B)	✓	✓	✓	✓	✓	
Finland	Yes (B)	✓		✓			✓
France	No	✓	✓	✓	✓	✓	
Germany	Yes (A)	✓	✓	✓			
Greece	No	✓	✓	✓			
Iceland	Yes (B)		✓				
Israel	Yes (B)	✓	✓			✓	
Ireland	Yes (A)	✓	✓	✓	✓	✓	
Italy	Yes (A)		✓	✓	✓		
Japan	No		✓		✓		
Korea	Yes (A)*	✓	✓	✓	✓	✓	
Lithuania	Yes (B)	✓	✓	✓	✓	✓	
Luxembourg	Yes (B)	✓	✓	✓	✓	✓	
Netherlands*	No			✓		✓	✓
New Zealand	No	✓	✓	✓	✓		
Norway	Yes (B)					✓	
Poland	No		✓		✓	✓	

	Requirements for schools to develop a plan or strategy concerning the use of digital resources for teaching (1)	Central initiatives to foster the effective use of digital resources for teaching					
		Official guidance on good practices for the use of digital resources for teaching	Official online platforms providing selected digital education content for teachers (2)	Official online platforms for teachers to share digital education content with peers	Publicly supported peer-to-peer training on the use of digital resources for teaching	Publicly supported initiatives on the use of AI for teaching	Other
Portugal	Yes (A)	✓	✓	✓	✓		
Slovak Republic	Yes (B)	✓	✓		✓		
Slovenia	No	✓	✓		✓		
Spain	Yes (A)	✓	✓	✓	✓	✓	
Sweden	No						✓
Other participants							
NSW (Australia)	Yes (B)	✓	✓				
Flemish Comm. (Bel)	No*		✓	✓	✓	✓	
French Comm. (Bel)	No	✓	✓	✓			
England (UK)	No		✓				
Northern Ireland (UK)	Yes (B)				✓	✓	
Scotland (UK)	No	✓	✓	✓	✓		
Wales (UK)	Yes (B)	✓	✓	✓	✓		
Partners							
Bulgaria	No		✓	✓			
Croatia	Yes (A)	✓	✓	✓	✓	✓	
Romania	Yes (B)	✓	✓	✓	✓		
Singapore	Yes (B)	✓	✓	✓	✓		

Note: (1) Requirements to include digital education in school-level strategies: Yes (A) = Yes, schools are required to develop plans or strategies concerning digital education; Yes (B) = Yes, digital education has to be covered by (more general) plans or strategies developed by the school.

(2) Digital education content refers to data that are produced, structured, distributed and presented in a way that aims to meet an educational objective. This includes education content in different formats, such as webpages, digital textbooks, movies, applications, etc.

Flemish Comm. (Bel): Schools are not required, but strongly encouraged and provided with tools to develop school-based ICT-policy plans. 70% of the Flemish schools have such a plan.

Korea: Only “research schools” and “leading schools” are required to develop school-level strategies / plans for digital education.

Netherlands: Following the implementation of an ongoing curriculum reform, it is expected that schools will be required to cover their approach to digital literacy in their school plans, starting in 2027.

Sources: Boeskens, L. and K. Meyer (2025^[2]), “Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age”, *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 2.3; Questions B.1 and B.9 of the Policy Survey on School Education in the Digital Age.

Annex Table B.2. Teachers' professional standards and appraisal frameworks

Coverage of teachers' use of digital resources in professional standards and formal appraisal or evaluation frameworks

ISCED level	The use of digital resources for teaching is included in teachers' professional standards or formal descriptions of their roles and responsibilities (1)			Teachers' work with digital resources is considered in formal appraisal/evaluation frameworks at the central level (2)		
	1	2	3	1	2	3
OECD countries						
Austria	a	a	a	a*	a*	a*
Chile	✓	✓	✓	✓	✓	✓
Colombia				✓	✓	✓
Denmark	✓	✓	✓	a	a	a
Estonia	✓	✓	✓	✓	✓	✓
Finland	a	a	a	a	a	a
France	✓	✓	✓	✓	✓	✓
Germany	a	a	a	m	m	m
Greece	✓	✓	✓	✓	✓	✓
Iceland	✓	✓	✓	a	a	a
Israel	✓	✓	✓			
Ireland	a	a	a	a	a	a
Italy	✓	✓	✓	a	a	a
Japan	a	a	a	a	a	a
Korea	✓	✓	✓			
Lithuania	✓	✓	✓	✓	✓	✓
Luxembourg	✓	✓	✓	a	a	a
Netherlands						
New Zealand				a	a	a
Norway	✓	✓	✓			
Poland	a	a	a			
Portugal	✓	✓	✓	a	a	a
Slovak Republic						
Slovenia						
Spain	✓	✓	✓	✓	✓	✓

	The use of digital resources for teaching is included in teachers' professional standards or formal descriptions of their roles and responsibilities (1)			Teachers' work with digital resources is considered in formal appraisal/evaluation frameworks at the central level (2)		
ISCED level	1	2	3	1	2	3
Sweden	a	a	a	a	a	a
Other participants						
NSW (Australia)	✓	✓	✓			
Flemish Comm. (Bel)				a	a	a
French Comm. (Bel)						
England (UK)				a	a	a
Northern Ireland (UK)	a	a	a			
Scotland (UK)	✓	✓	✓	a	a	a
Wales (UK)	✓	✓	✓	a	a	a
Partners						
Bulgaria	✓	✓	✓	✓	✓	✓
Croatia	✓	✓	✓	✓	✓	✓
Romania	✓	✓	✓	✓	✓	✓
Singapore	✓	✓	✓	✓	✓	✓

Note: (1) Jurisdictions without central professional standards or formal descriptions of teachers' roles and responsibilities were coded as "a" (not applicable); (2) Jurisdictions without central evaluation or appraisal framework for teachers were coded as "a" (not applicable).

Austria: The Federal Ministry of Education, Science and Research and some federal states recommend that school leaders use the digi.kompP framework (<https://www.virtuelle-ph.at/digikomp/>) to appraise the digital skills of their teachers, but the framework's use is not mandatory.

Sources: Boeskens, L. and K. Meyer (2025^[2]), "Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age", *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.1; Questions G.1 and G.2 of the Policy Survey on School Education in the Digital Age.

Annex Table B.3. Responsibilities for the use of digital resources and maintenance of digital devices in schools

	Dedicated staff positions with responsibilities related to the use of digital resources for teaching (1)															Staff responsible for maintaining digital resources in schools																	
	Teachers (Part-time role) (2)			Teachers (Full-time role)			Technical support staff (3)			Depending on school size or other school characteristics			Discretion of schools or local authorities			Other	Teachers (Part-time or full-time role)			Technical support staff (3)			External contractors			Depending on school size or other school characteristics			Discretion of schools or local authorities			Other	
ISCED level	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3			1	2	3	1	2	3	1	2	3	1	2	3	1	2		3
OECD countries																																	
Austria	A	A	A					B	B													B					*		*				
Chile*																																	
Colombia																																	
Denmark*																																	
Estonia																																	
Finland																																	
France	A	B	B																			B	B	B									
Germany																																*	
Greece																*																*	
Iceland														*	*	*																	
Israel	B	B	B				B	B	B																				*	*	*		
Ireland																																	
Italy																*																	
Japan							B	B	B																							*	
Korea							A*	A*	A*	*	*	*									A*	A*	A*			*	*	*					
Lithuania																																	
Luxembourg		A	A					A/B	A/B													A/B	A/B										
Netherlands																																	
New Zealand	A/B	A/B	A/B				A/B	A/B	A/B																								
Norway																																	
Poland	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m	m																	

	Dedicated staff positions with responsibilities related to the use of digital resources for teaching (1)															Staff responsible for maintaining digital resources in schools																
	Teachers (Part-time role) (2)			Teachers (Full-time role)			Technical support staff (3)			Depending on school size or other school characteristics			Discretion of schools or local authorities			Other	Teachers (Part-time or full-time role)			Technical support staff (3)			External contractors			Depending on school size or other school characteristics			Discretion of schools or local authorities			Other
ISCED level	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3		1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	
Portugal																																
Slovak Republic																																
Slovenia	A	A								A																		*	*			
Spain	A/B	A/B	A/B																													
Sweden																			A/B	A/B	A/B											
Other participants																																
NSW (Australia)														*	*	*													*	*	*	
Flemish Comm. (Bel)	B	B	B				A/B	A/B	A/B					*	*	*				A/B	A/B	A/B							*	*	*	
French Comm. (Bel)	A	A	A																													
England (UK)																																
Northern Ireland (UK)									A/B	A/B																						
Scotland (UK)																																
Wales (UK)																																
Partners																																
Bulgaria	A	A	A																													
Croatia							B	B	B										B	B	B											
Romania																																
Singapore*	A	A	A				A/B	A/B	A/B																							

Note: Cells shaded in blue = Yes; Cells shaded in white = No; a = There are no centrally defined staff positions with responsibilities related to the use of digital resources for teaching / for maintaining digital resources in schools at this level.

(1) Dedicated staff positions with responsibilities related to the use of digital resources for teaching include, for example, ICT co-ordinators supporting teachers with technical queries or helping them to integrate digital technologies into their practice; (2) A = Part-time roles associated with reduced teaching load; B = Part-time roles associated with additional remuneration; (3) A = Technical support staff based in one school; B = Technical support staff responsible for multiple schools.

Austria: For ISCED 1, municipalities are responsible for school maintenance, including ICT, and often hire external contractors for this. There are also teachers with reduced timetables providing IT support (e.g. for school management software) to several schools; For ISCED 2, responsibilities vary by school type (as at ISCED 1 for locally run schools, as at ISCED 3 for federal schools).

Chile: Although not a formal role, most schools (ISCED 1-3) have ICT co-ordinators with varying emphasis on technical and pedagogical aspects.

Denmark: Schools/municipalities decide whether to create roles for ICT advisors or co-ordinators, but part-time roles for teachers (with reduced teaching load) are a common practice.

Germany: Responsibility for ICT maintenance lies with the local school administration. Computer Science teachers usually serve as the “Head of the Computer Science Laboratory” and are responsible for all digital devices in the lab. They or other teachers with relevant knowledge usually also assume responsibility for the maintenance of other digital devices in the school.

Greece: School heads assign the responsibility for maintaining schools’ computer science labs to a CS teacher. In addition, there are dedicated staff positions at the district level with responsibilities related to the use of digital resources. In each Directorate of Education, “PLINET” (IT and New Technologies) technical support staff are assigned to assist all schools within their district. At the national level, the Panhellenic School Network (*Πανελλήνιο Σχολικό Δίκτυο*) technically supports teachers and schools in matters related to web applications and digital infrastructure.

Iceland: Assigning responsibilities for the use of digital resources in schools is at the discretion of local authorities. Responsibilities are commonly assigned to teachers in the school on a part-time basis (ISCED 1-3), to teachers with increased remuneration (ISCED 3) or to technical support staff shared between several schools (especially at ISCED 1 and 2).

Italy: As part of the Digital Plan, each school (ISCED 1-3) has a teacher serving as “digital animator”. Digital animators are part of a national network and have undergone dedicated training. With the school management team, they promote digital innovation in their school by organising trainings and workshops for staff and the wider school community and identifying sustainable digital solutions to integrate in their school.

Israel: The Ministry of Education provides schools with a budget for computer maintenance, which they can use to pay for a service supplier selected by the ministry or through the local authority.

Japan: The school founder is responsible for maintaining digital resources in schools (as defined in Article 2(2) of the School Education Act).

Korea: Tutors (ISCED 1) and IT Support Specialists (ISCED 1-3) are provided. Local governments can decide on their deployment based on the size and characteristics of each school and schools can take their own decisions on staff positions.

Slovenia: At ISCED 1 & 2, every school has at least one ICT co-ordinator responsible for the maintenance of digital infrastructure. However, schools can decide to outsource these tasks.

NSW (Australia): Schools decide how to assign responsibilities related to the use and the maintenance of digital resources (e.g. teachers or technical support staff). However, the assignment must adhere to established roles, responsibilities and relevant awards.

Flemish Community (Belgium): Schools decide how to allocate responsibilities for their digital resources but, as part of its digital strategy, the Flemish Community is emphasising the importance and potential of ICT teams comprised of different roles (as opposed to single ICT administrators). For further information: [Towards team-oriented ICT coordination at school](#).

French Community (Belgium): By law, schools can employ a *délégué-référent au numérique* (a member of the teaching staff that benefits from a reduction in teaching load to do digital-related tasks). Creating this position is at the discretion of schools but this is commonly done.

Singapore: All schools receive IT support from a headquarter-based ICT partner responsible for 1-2 school clusters. In addition, schools are encouraged to create positions for a Head of Department (ICT) and a school-based ICT Manager/Associate to support teachers in the use of digital resources.

Sources: Boeskens, L. and K. Meyer (2025^[2]), “Policies for the digital transformation of school education: Evidence from the Policy Survey on School Education in the Digital Age”, *OECD Education Working Papers*, No. 328, <https://doi.org/10.1787/464dab4d-en>, Annex Table 7.2; Questions G.3 and G.4 of the Policy Survey on School Education in the Digital Age.

Notes

¹ The following jurisdictions responded to the Policy Survey on School Education in the Digital Age: Austria, Chile, Colombia, Denmark, Estonia, Finland, France, Germany, Greece, Iceland, Israel, Ireland, Italy, Japan, Korea, Lithuania, Luxembourg, Netherlands, New Zealand, Norway, Poland, Portugal, the Slovak Republic, Slovenia, Spain, Sweden, New South Wales (NSW) in Australia, the Flemish Community of Belgium, the French Community of Belgium, England (United Kingdom), Northern Ireland (UK), Scotland (UK), Wales (UK), Bulgaria, Croatia, Romania and Singapore. Throughout this document, the terms “jurisdictions” and “systems” are used interchangeably to refer to the respondents to the Policy Survey.

² In some of the other education systems, the teacher evaluation process is organised at the regional or local level (e.g. in Spain and New South Wales [Australia]) or at the school level (e.g. in Portugal and Sweden).

³ See Annex Table 9.2. in Boeskens and Meyer (2025^[2]) for an overview of the criteria and processes behind teacher career structures.

The digital transformation of education presents both opportunities and challenges for teaching practices and teachers' working conditions. While many educators have strengthened their digital competencies in recent years, persistent technical and pedagogical barriers continue to hinder the effective and deliberate integration of digital tools in the classroom. This policy paper examines how teacher policies can be adapted to better support the profession in navigating this transition. Drawing on insights from PISA 2022 and comparative data from the Policy Survey on School Education in the Digital Age, it addresses the following questions: What forms of central guidance can support teachers' effective use of digital resources? To what extent do professional standards and appraisal frameworks reflect the evolving digital education landscape? How can technology alleviate teachers' workload and optimise their use of time? Do staff roles and career structures need to be reshaped to enhance schools' digital capacity? Which mechanisms can ensure that teachers receive ongoing technical assistance and support in maintaining digital resources? The paper offers evidence-informed insights for policy makers seeking to strengthen teacher policies to build resilient, future-ready digital education systems.

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